

Problem 1. Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17.0/24. Also suppose that Subnet 1 is required to support up to 62 interfaces, and is already assigned the subnet address block 223.1.17.0/26.

a) If subnet 2 is to support up to 106 interfaces, what will be the allocated address block in the form of "IP address/subnet mask" in decimal format, such as a.b.c.d/x?

b) If subnet 3 is to support up to 14 interfaces in the remaining addresses, what will be the allocated address block in the form of "IP address/subnet mask" in decimal format, such as a.b.c.d/x?

Hint: there shall be no overlap between the assigned network addresses among the three subnets. Otherwise there could be IP address conflict within the router.

Problem 2. Why are different inter-AS and intra-AS protocols used in the Internet?

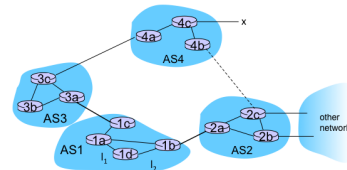
Problem 3. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eGBP and iGBP are used for the inter-AS routing protocol. Initially suppose there is no physical link between AS2 and AS4.

a) Router 3c learns about prefix x from which routing protocol: OSPF, RIP, eGBP, or iGBP?

b) Router 3a learns about x from which routing protocol?

c) Router 1c learns about x from which routing protocol?

d) Router 1d learns about x from which routing protocol?



A **network** is a group of connected devices (like computers, phones, or printers) that can communicate with each other directly.

In IP addressing (IPv4), a network is defined by:

- ❑ A **network address** (the "base" IP of the group, e.g., 223.1.17.0).
- ❑ A **subnet mask** (e.g., /24), which tells devices: "These bits are for the network ID, and these are for individual devices."

Key Idea: Devices use network address to know if traffic is "local or needs to go to a router.

Example: In 223.1.17.0/24:

- ❑ Network address: 223.1.17.0 (all host bits set to 0).
- ❑ This covers addresses from 223.1.17.0 to 223.1.17.255 (256 total spots, but 2 reserved).
- ❑ Devices on this network can "talk" without leaving the group.

A **host** is any single device connected to a network that can send or receive data. Examples: your laptop, smartphone, server, or even a router's interface.

Each host gets a unique IP address within its network (e.g., 223.1.17.5)

- ❑ The IP is split: Network portion (fixed for the group) + Host portion (unique to the device).

Key Idea: Hosts are the "end users" of the network. Without hosts, a network is just empty space.

Reserves in a Subnet:

- ❑ First address: Network ID (not for hosts).
- ❑ Last address: Broadcast (for "shouting" to all hosts, not for a single device).
- ❑ Middle ones: Usable for hosts (e.g., in /26 subnet: 62 usable hosts).

What is Subnetting?

In Simple Terms: Subnetting breaks a large IP network into smaller, more manageable sub-networks (subnets). Each subnet acts like its own mini-network with its own range of addresses.

Why It Exists: A huge network (e.g., 256 addresses) might be too big for one office floor—it's wasteful and hard to manage. Subnetting lets you create:

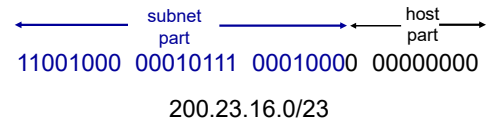
- ❑ Smaller groups for different departments (e.g., HR gets 50 addresses, IT gets 100).
- ❑ Better security (traffic stays local to the subnet).
- ❑ Efficiency (routers handle less "broadcast" noise).

Real-World Analogy: Your city's zip code (big network) gets split into neighborhoods (subnets) for mail sorting—each neighborhood has its own block of addresses.

CIDR notation

CIDR: Classless InterDomain Routing

- subnet portion of address of arbitrary length
- address format: **a.b.c.d/x**, where x is # bits in subnet portion of address



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Network Layer 4-4

IP addresses: how to get one?

Key Idea: No one starts from scratch. IP addresses are handed out in a pyramid: Global registries → ISPs → You/Your Organization.

Q: how does *network* get subnet part of IP addr?

A: gets allocated portion of its provider ISP's address space

ISP's block	11001000 00010111 00010000 00000000	200.23.16.0/20
Organization 0	11001000 00010111 00010000 00000000	200.23.16.0/23
Organization 1	11001000 00010111 00010010 00000000	200.23.18.0/23
Organization 2	11001000 00010111 00010100 00000000	200.23.20.0/23
...		
Organization 7	11001000 00010111 00011110 00000000	200.23.30.0/23

Problem 1

You only need to care about how to allocate the last 8 bits, no overlap among subnet1, subnet2, and subnet3.

Subnet1: 223.1.17.0/26	→	00 000000	$2^6 = 64 > 62$
Subnet2: 223.1.17.128/25	→	1 0000000	$2^7 = 128 > 106$
Subnet3: 223.1.17.64/28	→	0100 0000	$2^4 = 16 > 14$

223.1.17.0/26	11011111	00000001	00010001	00000000
223.1.17.128/25	11011111	00000001	00010001	10000000
223.1.17.64/28	11011111	00000001	00010001	01000000

Other choices for subnet3:

223.1.17.80/28	→	0101 0000	$2^4 = 16 > 14$
223.1.17.96/28	→	0110 0000	$2^4 = 16 > 14$
223.1.17.112/28	→	0111 0000	$2^4 = 16 > 14$

Network Layer 4-5

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Hierarchical routing

our routing study thus far - idealization

- ❖ all routers identical
- ❖ network “flat”
- ... *not* true in practice

scale: with 600 million destinations:
❖ can't store all dest's in routing tables!
❖ routing table exchange would swamp links!

administrative autonomy
❖ internet = network of networks
❖ each network admin may want to control routing in its own network

Network Layer 4-7

Hierarchical routing

- ❖ aggregate routers into regions, “**autonomous systems**” (AS)
- ❖ routers in same AS run same routing protocol
 - “**intra-AS**” routing protocol
 - routers in different AS can run different intra-AS routing protocol

gateway router:

- ❖ at “edge” of its own AS
- ❖ has link to router in another AS

Network Layer 4-8

Why different Intra-, Inter-AS routing ?

policy:

- ❖ inter-AS: admin wants control over how its traffic routed, who routes through its net.
- ❖ intra-AS: single admin, so no policy decisions needed

scale:

- ❖ hierarchical routing saves table size, reduced update traffic

performance:

- ❖ intra-AS: can focus on performance
- ❖ inter-AS: policy may dominate over performance

Network Layer 4-9

Prob. 2 (Offline Reading)

- Policy: **Among ASs, policy issues dominate**. It may well be important that traffic originating in a given AS not be able to pass through another specific AS. Similarly, a given AS may want to control what transit traffic it carries between other ASs. **Within an AS**, everything is nominally under the same administrative control and thus **policy issues a much less important** role in choosing routes within AS.
- Scale: The ability of a routing algorithm and its data structures to scale to handle routing to/among large numbers of networks is a critical issue in inter-AS routing. Within an AS, scalability is less of a concern. For one thing, if **a single administrative domain becomes too large**, it is always possible to **divide it into two ASs** and perform inter-AS routing between the **two new ASs**.
- Performance: Because **inter-AS routing** is so policy oriented, the **quality** (for example, performance) of the routes used is often of **secondary concern** (that is, a longer or more costly route that satisfies certain policy criteria may well be taken over a route that is shorter but does not meet that criteria). Indeed, we saw that **among ASs, there is not even the notion of cost (other than AS hop count)** associated with routes. **Within a single AS**, however, such **policy concerns are of less importance**, allowing routing to focus more on the level of performance realized on a route.

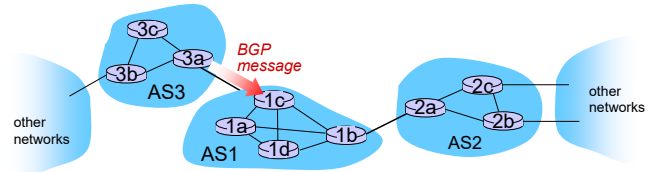
Internet inter-AS routing: BGP

- ❖ **BGP (Border Gateway Protocol):** the de facto inter-domain routing protocol
 - “glue that holds the Internet together”
- ❖ BGP provides each AS a means to:
 - **eBGP:** obtain subnet reachability information from neighboring ASs.
 - **iBGP:** propagate reachability information to all AS-internal routers.
 - determine “good” routes to other networks based on reachability information and policy.
- ❖ allows subnet to advertise its existence to rest of Internet: “*I am here*”

Network Layer 4-11

BGP basics

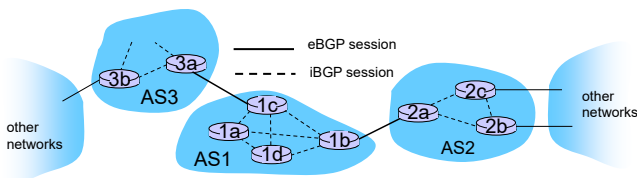
- ❖ **BGP session:** two BGP routers (“peers”) exchange BGP messages:
 - advertising *paths* to different destination network prefixes (“path vector” protocol)
 - exchanged over semi-permanent TCP connections
- ❖ when AS3 advertises a prefix to AS1:
 - AS3 *promises* it will forward datagrams towards that prefix
 - AS3 can aggregate prefixes in its advertisement



Network Layer 4-12

BGP basics: distributing path information

- ❖ using eBGP session between 3a and 1c, AS3 sends prefix reachability info to AS1.
 - 1c can then use iBGP to distribute new prefix info to all routers in AS1
 - 1b can then re-advertise new reachability info to AS2 over 1b-to-2a eBGP session
- ❖ when router learns of new prefix, it creates entry for prefix in its forwarding table.



Network Layer 4-13

Problem 3

- A) eBGP
- B) iBGP
- C) eBGP
- D) iBGP